Exercises and solutions

All twenty-four lectures · 120 questions, each answered

How to use this

These are the five questions set at the end of each lecture's deck, in the style of the written paper, with the solutions that appear on the following lecture's deck. Everything here is on the slides already; this is the same material in one place, for revision.

They are also the oral. If you sit the optional oral, three of these 120 are drawn in front of you and you answer them aloud, marked against the same scheme printed here with each solution. There is no second, hidden bank: this is it.

Each question is answerable from **its own lecture and the ones before it**, and from nothing else — that is a rule the course checks mechanically, not an aspiration. So if a question seems to need something you have not met, the fault is ours: say so.

The mark in brackets is what the question would be worth on the paper, and it is the best guide to how long an answer should be. Four marks is a short paragraph, not a page.

Contents

Lecture 1 What machine learning is, and how we will work

Ch 1–2

1. A colleague reports that their model reaches an RMSE of 48,000 on the California housing data, and that they chose the model by trying six of them and keeping the best. State, in one sentence, what that number is an estimate of — and what it is *not* an estimate of. **[4]**

Solution

It estimates the best-of-six score on that particular split; it does not estimate the error on new data.

- Choosing the minimum of six numbers measured on the same data makes that minimum a *selection*, not a measurement
- The quantity a client cares about — error on districts nobody has seen — needs data that took part in no choice

2. The median income column is capped at 15. Give one consequence for the *model* and one for the *evaluation*, and say which of the two you would raise with the client first. **[5]**

Solution

Model: it can never learn what happens above the cap. Evaluation: the test set is capped too, so the error is silent about exactly those districts.

- The cap is in both halves of the split, so no held-out number reveals it
- Raise the evaluation one first: the model can be retrained, but a metric that cannot see the failure will not tell you to

3. You are given a stratified split on income category and a naive random split of the same data. Both report similar RMSE. Does that mean the stratification was unnecessary? Answer yes or no, with a reason. **[4]**

Solution

No.

- Similar RMSE on *one* draw says nothing about the spread over draws, and the stratification is there to reduce that spread
- The failure a naive split causes is occasional and large, which is precisely what a single comparison cannot show

4. The brief says the output feeds a downstream system that expects a price. Name the one thing this fact settles about the problem framing, and one thing it leaves open. **[4]**

Solution

It settles that this is supervised regression, not classification. It leaves the metric open.

- 'A price' fixes the type of the output and nothing about how error is weighed
- Whether being wrong by 50,000 on a cheap district matters as much as on an expensive one is a question for the client, not the data

5. In the working loop — specify, generate, read, test, verify — exactly one step is done by the assistant. Name it, and say what goes wrong if a student treats a second step as the assistant's. **[3]**

Solution

Generate. If reading is also delegated, nothing checks that the code does what the specification asked for.

- The specification is the student's claim about what the code must do
- Only a reader who holds that claim can notice the code meeting a different one

Lecture 2 The end-to-end project

Ch 2

1. The normal equation is $\hat{\theta} = (X^\top X)^{-1} X^\top y$. Your design matrix has a column equal to the sum of two others. State what happens to $X^\top X$, and what `np.linalg.pinv` returns instead. **[5]**

Solution

$X^\top X$ is singular; the pseudoinverse returns the minimum-norm least-squares solution.

- A linear dependence among columns makes X rank-deficient, so $X^\top X$ has a zero eigenvalue and no inverse
- The fitted *values* are still unique — the projection onto the column space is — only the coefficients are not

2. A pipeline scales the features and then fits a ridge model, and is passed whole to `cross_val_score`. Explain what would go wrong if the scaling were done once, before the call. **[5]**

Solution

The scaler would be fitted on every fold's validation rows, so each fold's score would be optimistic.

- A scaler is a fitted object: its mean and variance are parameters learned from data
- Fitting it before the split lets validation rows influence the transformation applied to themselves

3. You report a 95% confidence interval for the test RMSE. A colleague says the interval proves the model is better than the baseline. Give the one question you would ask before agreeing. **[4]**

Solution

Was the baseline scored on the same test rows?

- An interval describes the uncertainty of *one* estimate, not the difference between two
- Comparing two systems needs their difference's spread, which needs them measured on the same rows

4. k-fold cross-validation reports a mean of 0.71 with folds $[0.52, 0.79, 0.75, 0.74, 0.75]$. What do you conclude, and what would you do next? **[5]**

Solution

One fold is an outlier; investigate it rather than reporting the mean.

- Four folds agree closely and one does not, which is a property of the data in that fold, not noise around a common value
- The mean of a bimodal set of scores describes none of them

5. Explain, in two sentences, why the test set may be looked at exactly once, and what you may legitimately do if the number disappoints. **[4]**

Solution

Looking at it more than once makes it a validation set. If it disappoints, you may report it — not tune against it.

- Every decision taken after seeing the test score is a decision the score no longer measures honestly
- The permitted response is a new experiment on new held-out data, or an honest report of what was found

Lecture 3 Classification and its metrics

Ch 3

1. On the MNIST 5-detector, a classifier that never fires scores 90.96% accuracy on the training set. State what that number is a measurement of. **[3]**

Solution

The share of the data that is not a 5.

- With no positive prediction, accuracy is exactly the negative class's base rate
- It measures the data, not the model — which is why it is the anchor every other number is read against

2. Prove or disprove: as the decision threshold is lowered, precision increases monotonically. **[5]**

Solution

Disprove — precision is not monotone in the threshold.

- Lowering the threshold admits one instance at a time; recall cannot fall, but precision rises or falls depending on whether that instance is a true or a false positive
- A single false positive admitted at a high-precision point lowers it, and a later true positive can raise it again

3. A client asks for "90% precision". Explain why that is not yet a specification, and give the one thing you would add. **[4]**

Solution

It names no recall, and precision alone is met by predicting almost nothing. Add the recall it must hold at.

- A classifier firing once, correctly, has precision 1.0 and is useless
- An operating point is a pair, and the threshold that achieves it is a decision someone has to own

4. Two models have the same ROC AUC. One is far better on a precision–recall curve. What does that tell you about the problem? **[4]**

Solution

The positive class is rare.

- ROC uses the false-positive *rate*, whose denominator is the large negative class, so many false positives barely move it
- PR uses precision, whose denominator is what the model predicted, so the same false positives dominate it

5. The confusion matrix of the never-fires classifier has two zero cells. Name them, and say which single metric would have exposed the model immediately. **[4]**

Solution

True positives and false positives are zero; recall exposes it, at 0.0.

- Nothing was predicted positive, so both positive-prediction cells are empty
- Recall divides by the actual positives, so it is 0 however rare they are

Lecture 4 Training models

Ch 4

1. Gradient descent on the same data, with the features unscaled, takes many more steps to converge than on scaled features. Explain why, in terms of the shape of the cost surface. **[5]**

Solution

Unscaled features make the contours elongated, so the gradient points across the valley rather than along it.

- The Hessian is $X^\top X$, so the curvature along a feature goes as the *square* of its scale — multiply a column by ten and its curvature rises a hundredfold, which is how one column comes to dominate
- The learning rate must then suit the steepest direction, which makes it far too small for the shallow one

2. Your logistic regression on one-hot encoded columns produces coefficients that look wrong, and the design matrix has rank six less than its column count. State the cause and the standard repair. **[5]**

Solution

Five come from the encoder, one from you. The repair is both: drop one level per categorical, and delete the engineered column.

- Each one-hot block sums to 1 in every row, and so does the intercept column — five categoricals, five exact dependencies
- The sixth is `FamilySize = SibSp + Parch + 1`, an identity rather than a correlation. Dropping a level alone leaves rank 23 of 24, because the dependency you wrote yourself survives it

3. A model reaches 0.80 accuracy and is *badly calibrated*. Explain what that means, and name a decision it would make wrongly that accuracy cannot see. **[4]**

Solution

Its predicted probabilities do not match observed frequencies; any decision using a cost-weighted threshold is then wrong.

- Accuracy reads only the arg-max, so it is blind to the probability that produced it
- Choosing a threshold from expected cost requires the probability to mean what it says

4. Why does the log loss have no closed-form minimiser, when squared error does? **[4]**

Solution

Its gradient is not linear in the parameters, so setting it to zero gives no solvable system.

- Squared error is quadratic in θ, so its gradient is affine and the stationary condition is a linear system
- The sigmoid makes the log-loss gradient transcendental in θ

5. State the update rule for batch gradient descent, and say what changes in it for stochastic gradient descent — and what does not. **[4]**

Solution

$\theta \leftarrow \theta - \eta \nabla J$. Only the set the gradient is computed over changes; the rule does not.

- SGD estimates the same gradient from one instance rather than all of them
- The estimate is unbiased and noisy, which is why the step size usually has to decay

Lecture 5 Regularisation and the bias–variance trade-off

Ch 4

1. Write the bias–variance decomposition of the expected squared error, and name the one term no model can reduce. **[4]**

Solution

$\mathbb{E}[(y - \hat{f})^2] = \text{bias}^2 + \text{variance} + \sigma^2$; the noise σ^2 cannot be reduced.

- Two passengers with identical features and different outcomes put a floor under any predictor
- A model reporting error below that floor has been measured wrongly

2. A learning curve shows training and validation error both high and close together, and flat as data is added. Diagnose it, and say whether more data would help. **[5]**

Solution

High bias. More data would not help.

- The two curves have already met, so the gap — the variance — is small
- The remaining error is the model's inability to represent the function, which more rows do not change

3. Ridge adds αI to $X^\top X$ before inverting. Give the numerical consequence, in terms of the condition number. **[4]**

Solution

It raises every eigenvalue by α, so the condition number falls and the inverse is stable.

- Near-collinear columns give eigenvalues near zero, which the inverse amplifies
- The bound on the solution's sensitivity improves as α grows — at the cost of bias

4. Why must features be scaled before ridge or lasso, when they need not be for ordinary least squares? **[4]**

Solution

The penalty is on the coefficients, and a coefficient's size depends on its feature's units.

- Least squares is equivariant to rescaling a column: the coefficient absorbs it
- A penalised fit is not, so an unscaled feature is penalised in proportion to the units someone chose

5. You tune α over a grid using cross-validation, then report the best cross-validated score as your estimate of future performance. Name the error, and quantify its direction. **[5]**

Solution

The best-of-grid score is optimistic: it is a minimum over the same folds used to choose.

- Selection over k candidates on the same data biases the winner downward in error
- The honest number comes from data that took part in no choice — a held-out test set, scored once

Lecture 6 Decision trees

Ch 5

1. Gini and entropy give nearly the same trees on CoverType. State the property they share that explains it, and name a case where they can differ. **[4]**

Solution

Both are strictly concave with a maximum at the uniform distribution and zero at a pure node; they differ only where two splits are nearly tied.

- Any impurity with those properties ranks most candidate splits identically
- They can part only on a pair of candidate splits whose children are incomparable in that ordering and whose weighted impurities differ by less than 0.055 — a close call deep in the tree, never at the root

2. A decision tree is described as 'nonparametric'. Explain what that means here, and what it implies about the need for hyperparameters. **[4]**

Solution

Its structure is not fixed before fitting, so it will grow to fit the data exactly unless constrained.

- The parameter count is decided by the data, not by the model class
- Constraints such as depth or minimum leaf size are what stop it memorising

3. The brief allows at most eight conditions per decision. You measure the number of conditions as `decision_path().sum() - 1`. Explain the -1. **[3]**

Solution

The decision path counts the nodes visited, including the leaf, and the leaf applies no condition.

- A path of depth d visits $d + 1$ nodes
- Without the subtraction the count exceeds the tree's own depth, which is what the assert catches

4. Twenty trees are refit on 90% subsamples of one training set, so any two of them share about 80% of their rows — and they still give visibly different trees. Does this make the model unreliable? Answer, with a reason. **[5]**

Solution

Yes, in the way that matters. The accuracy is stable to 0.25 points while 9.1% of individual predictions change — a stable metric is not evidence of a stable model.

- Accuracy averages over 12,000 patches, and an average hides substitutions: a tree that loses one patch and gains another scores the same
- The instability is deep in the tree, not at the root — the root feature is Elevation in all twenty refits — so the single decision you would be asked to explain is the part that does not reproduce

5. Per-class recall shows one class far below the rest. Give two responses that do not involve changing the model, and say which you would try first. **[4]**

Solution

Report per-class recall beside the headline, and take the question back to the client. Report first.

- A single accuracy averages the class away; per-class recall costs nothing and makes the failure visible, which is the only response available before anyone has decided anything
- Whether that class is worth other classes' recall is the agency's decision, not a machine learning one — and reweighting, the obvious next step, is a different model answering a different question

Lecture 7 Ensembles and random forests

Ch 6

1. For n predictors each of variance σ^2 and pairwise correlation ρ, the variance of their average is $\rho\sigma^2 + \frac{1-\rho}{n}\sigma^2$. State the limit as $n \to \infty$, and what it means for ensembles. **[5]**

Solution

It tends to $\rho\sigma^2$ — correlation, not ensemble size, sets the floor.

- The second term vanishes; the first does not depend on n at all
- Adding members past a point buys nothing, which is why the design effort goes into *decorrelating* them

2. Extra-trees are usually faster to fit than a random forest and often no less accurate. Give the mechanism, in terms of the formula above. **[4]**

Solution

Speed, not a lower floor. Not searching for the best threshold is what makes them fast; measured against the forest their ρ is *higher* (0.067 against 0.043), and they stay competitive because the two mechanisms are substitutes rather than additions.

- Extra-trees without a bootstrap land at $\rho = 0.067$, near bagging; with one they land at 0.043, exactly where the forest is — random thresholds substitute for the bootstrap rather than adding to it
- Their individual members are *better*, not worse — 77.8% against 73.6% — because a random threshold on every feature beats an optimal threshold on a subsample of them

3. You set `bootstrap=False` and `oob_score=True`. State what happens and why. **[3]**

Solution

It raises: with no bootstrap there are no out-of-bag rows.

- Out-of-bag scoring uses the rows a member did not sample
- Without sampling, every member sees every row and the estimate is undefined

4. Impurity-based feature importance ranks a random decoy column above several real ones. Explain how that can happen, and name the measure you would use instead. **[5]**

Solution

A high-cardinality column offers more split points, so it accumulates impurity decrease by chance. Use permutation importance.

- Impurity importance rewards a column for being *splittable*, not for being informative
- Permutation importance measures the score lost when the column is shuffled, which a decoy cannot fake

5. A forest improves accuracy over one tree by far less than the variance reduction would suggest. Is this a contradiction? Answer with a reason. **[4]**

Solution

No — variance is only one term of the error.

- Averaging reduces variance and leaves bias untouched
- If the remaining error is mostly bias or noise, removing variance moves the accuracy very little

Lecture 8 Dimensionality reduction and unsupervised learning

Ch 7–8

1. PCA is derived by maximising $\|XW\|_F^2$ subject to $W^\top W = I$. State why the data must be centred first, and what the first component becomes if it is not. **[5]**

Solution

Without centring the leading direction points at the mean, not at the direction of greatest variance.

- The objective measures distance from the origin, and only after centring is that the same

as spread
- On the Olivetti faces the uncentred first component is the mean face

2. The Johnson–Lindenstrauss bound, evaluated for 400 images at $\varepsilon = 0.1$, asks for more dimensions than the data has. What do you conclude about the bound, and about the method? **[5]**

Solution

The bound is vacuous here; the method still works.
- JL is a worst-case guarantee over all point sets, and this one is not worst case
- A bound that does not bind is not evidence against the technique — measure the distortion instead

3. A pipeline fits PCA on the whole dataset and then splits. Name what leaked, and say whether the leak grows or shrinks as the corpus grows. **[4]**

Solution

The components leaked; the damage shrinks as the corpus grows.
- The principal directions are fitted parameters, so fitting them on test rows lets those rows shape their own representation
- With more rows each individual test row moves the components less

4. Inertia always falls as k rises, so it cannot choose k. State what silhouette adds, and one situation where it also fails. **[5]**

Solution

Silhouette compares within-cluster to nearest-other-cluster distance, so it can fall. It fails when the clusters are not compact and separated.
- Inertia is monotone by construction: more centres can only shorten distances
- Silhouette assumes the geometry k-means assumes, so elongated or nested clusters defeat both

5. You have 4,096-dimensional face vectors and are told two faces are "close". Explain why that alone does not mean they are the same person. **[4]**

Solution

Distance is not identity — lighting and pose move a face further than identity does.
- Pixel distance is dominated by whatever varies most, and that is illumination
- The metric has to be one under which the thing you care about is the thing that varies

Lecture 9 Neural networks, from the perceptron up

Ch 9

1. A multilayer perceptron with the activation removed is written as $W_3W_2W_1x$. State what function class it can represent, and why depth then buys nothing. **[4]**

Solution

Only linear maps — the product of matrices is a matrix.

- Composition of linear maps is linear, whatever the depth
- The nonlinearity is what makes an extra layer a new function rather than a re-parameterisation

2. Give the parameter count of a dense layer from m inputs to n units, and evaluate it for the first layer of a network on 28×28 images with 300 units. **[4]**

Solution

$mn + n$; here $784 \times 300 + 300 = 235{,}500$.

- One weight per input-output pair, plus one bias per unit
- The first layer of an image network usually dominates the count, because m is the pixel count

3. A single TLU cannot represent XOR. State the property of XOR that prevents it, and the smallest change to the network that fixes it. **[4]**

Solution

XOR is not linearly separable; one hidden layer fixes it.

- A TLU's decision boundary is a hyperplane, and no line separates the two XOR classes
- A hidden layer can re-represent the inputs so that the classes become separable in the new coordinates

4. Fashion-MNIST is exactly balanced across ten classes. State the trivial baseline's accuracy, and why balance is what makes accuracy defensible here. **[3]**

Solution

10%. With equal class sizes and equal costs, accuracy is not dominated by one class.

- The majority-class predictor scores the largest class share, which is one tenth
- Lecture 3's objection to accuracy was imbalance, and it does not apply

5. An architecture sweep finds the best hidden-layer size on this dataset. State what that result does and does not transfer to. **[4]**

Solution

Very little. It holds for this task at the training-set size you swept at, and not even for the same dataset at full size: the architecture the sweep ranked third of five is the one that wins on all 55,000 images.

- A deeper network needs more data to pay for its extra parameters, so a sweep run on a fifth of the data systematically prefers the shallow ones — and never finds out
- It is a measurement of one sweep at one size, not a rule. Reporting it as a general recommendation is the error

Lecture 10 PyTorch

Ch 10

1. Reverse-mode automatic differentiation costs one sweep for all partial derivatives; forward mode costs one per input. State the condition on the function's shape that makes reverse mode the right choice. **[4]**

Solution

Many inputs, one output — which is exactly a loss.

- Forward mode's cost scales with the number of inputs, reverse mode's with the number of outputs
- A neural network has millions of parameters and one scalar loss

2. Explain why the loss must be a scalar for `.backward()` to be called without arguments. **[3]**

Solution

The reverse sweep is seeded with $\partial L/\partial L = 1$, which only exists for a scalar.

- For a vector output there is no single derivative to seed with
- PyTorch then requires an explicit vector to weight the outputs by

3. The five lines of the training loop are reordered so that `opt.step()` comes before `loss.backward()`. State what the student observes. **[4]**

Solution

Nothing raises, and the model stays at chance.

- The step is taken with the gradients zeroed at the top of the iteration
- The gradients computed afterwards are cleared before they are ever used

4. A network with dropout is evaluated without calling `model.eval()`. Describe the symptom precisely, and the one-line diagnostic that catches it. **[5]**

Solution

The score wobbles between identical calls. Evaluate twice and assert the two agree.

- Dropout keeps sampling masks in training mode, so the metric is a random variable
- A deterministic function of fixed weights and fixed data does not change between calls

5. GPUs did not help at small model sizes in the measured comparison. Give the reason, and the quantity that decides where the crossover sits. **[4]**

Solution

Transfer and launch overhead dominates; the crossover is set by arithmetic per byte moved.

- A small matrix product finishes faster than the cost of getting it there
- The device pays once the work per transferred byte is large enough

Lecture 11 Training deep networks

Ch 11

1. One layer multiplies the standard deviation of the backward signal by $\rho = \sqrt{n_{\text{out}}\,\mathrm{Var}(w)\,\mathbb{E}[\varphi'^2]}$. State what L layers do to it, and why only $\rho = 1$ survives depth. **[5]**

Solution

They multiply it by ρ^{L-1} — $L-1$ steps between the first gradient and the last — so anything but 1 vanishes or explodes geometrically.

- A constant applied L times is exponential in L
- There is no regime where a repeated constant other than 1 is safe

2. Preserving the forward signal asks for $\mathrm{Var}(w) = 1/n_{\mathrm{in}}$ and preserving the gradient asks for $1/n_{\mathrm{out}}$. State when the two agree, and what Glorot does when they do not. **[4]**

Solution

They agree only when $n_{\mathrm{in}} = n_{\mathrm{out}}$; Glorot takes the harmonic mean, $2/(n_{\mathrm{in}} + n_{\mathrm{out}})$.

- Any layer that changes width cannot satisfy both
- The compromise accepts a small error in each direction rather than a large one in either

3. He initialisation doubles the weight variance for ReLU. Derive the factor of two in one line. **[4]**

Solution

ReLU zeroes half of a symmetric zero-mean input, so $\mathbb{E}[\mathrm{ReLU}(z)^2] = \frac{1}{2}\mathbb{E}[z^2]$.

- Exactly half the variance is discarded at each layer
- Doubling $\mathrm{Var}(w)$ puts it back

4. A gradient profile that is a straight line on a log axis tells you something a curved one does not. State what, and why it identifies the cause. **[4]**

Solution

The attenuation is the same factor at every layer, so it is the initialisation rather than one bad layer.

- A constant ratio per layer is a straight line in log space
- A single broken layer would show a step, not a slope

5. Gradient clipping is applied to a network whose gradients are fifteen orders of magnitude too small. Predict the effect, and say what the attempt reveals about the diagnosis. **[4]**

Solution

No effect — clipping bounds gradients that are too large.

- The threshold is never reached, so nothing is rescaled
- Reaching for it means the failure mode was never measured

Lecture 12 Convolutional networks

Ch 12

1. A dense layer mapping a $3 \times 128 \times 128$ input to a $32 \times 128 \times 128$ output has $n_{\mathrm{in}} n_{\mathrm{out}}$ weights. Give the convolutional count for a 7×7 kernel, and say which symbols are absent from it. **[5]**

Solution

$32 \times 3 \times 7 \times 7 = 4{,}704$. Neither H nor W appears.

- One kernel per output channel, reused at every position
- The image size is genuinely absent — that is what weight sharing means

2. State the difference between equivariance and invariance, and say which of the two segmentation cannot afford. **[4]**

Solution

Equivariance: $f(Tx) = Tf(x)$. Invariance: $f(Tx) = f(x)$. Segmentation cannot afford invariance.

- Invariance discards the position, and the answer to a segmentation question *is* the position
- Classification wants invariance for the same reason

3. Convolution is equivariant to translation on the interior of an image but not at the border. Give the reason. **[3]**

Solution

Zero padding invents input that was not there, so a shift moves real content into invented content.

- The identity assumes the operator sees the same neighbourhood before and after the shift
- At the edge it does not, so the equality is an interior statement

4. A training run fails with out-of-memory. State which of parameters and activations usually dominates, and give the first thing to change. **[4]**

Solution

Activations dominate; halve the batch.

- Activation memory is linear in the batch size and parameter memory does not depend on it at all
- Reducing the parameter count is near the bottom of the list

5. Parameters and activations are anti-correlated across a convolutional stack. State where each is concentrated, and what that means for any intuition carried over from dense networks. **[4]**

Solution

Activations are largest near the input, parameters near the output. Dense intuition points the wrong way.

- Early layers have large maps and tiny kernels; late layers the reverse
- In a dense network both grow together, so the habit transfers badly

Lecture 13 Transfer learning

Ch 12

1. State the order in which a pretrained network's head must be replaced and its body frozen, and what goes wrong in the other order. **[4]**

Solution

Freeze first, then replace. In the other order the new head is frozen too.

- A newly constructed module has `requires_grad=True`, so the freezing loop would turn it off
- The symptom is a loss that never moves, with nothing raising

2. Fine-tuning uses 10^{-4} on the pretrained block and 10^{-3} on the new head. Justify the difference in one sentence. **[4]**

Solution

The block already encodes something and the head starts from noise; one rate cannot suit both.

- A large step destroys what the block knows
- A small step leaves the random head untrained

3. Freezing a batch-norm layer's weights does not freeze the layer. State what still changes, and the call that stops it. **[4]**

Solution

The running statistics are buffers, updated in the forward pass; `.eval()` stops them.

- `requires_grad=False` governs gradients, not buffers
- A frozen feature extractor left in train mode still drifts

4. An augmented validation loader is used to select the epoch to keep. State the two things this costs, and the one-line test that detects it. **[5]**

Solution

The score becomes a random variable and the chosen epoch changes. Evaluate twice and require identical answers.

- Augmentation makes the metric depend on the evaluator's seed
- A deterministic function of fixed weights and fixed data does not wobble

5. A frozen-backbone probe is both more accurate and faster than training from scratch on 1,020 images. Explain why both, and name the cost that must still be counted. **[5]**

Solution

The backbone is a fitted function reused for free; the feature extraction pass must still be counted in the total.

- Only the head is fitted, so the variance problem moves to a dataset large enough to absorb it
- The probe is not free just because the head is

Lecture 14 Detection and segmentation

Ch 12

1. Two boxes are disjoint and moved further apart. State what IoU and its derivative do, and why no learning rate repairs it. **[5]**

Solution

Both are identically zero; a constant has no descent direction.

- Past contact the intersection is exactly zero for every separation
- The gradient is absent rather than small, so no optimiser can act on it

2. An IoU implementation without a clamp is tested on overlapping boxes only and passes. Give the input that breaks it, and the property test that would have caught it. **[5]**

Solution

Boxes separated on both axes. Assert the result lies in $[0, 1]$ and is zero exactly when the boxes are disjoint.

- Two negative differences multiply to a positive 'intersection'
- A range check alone is necessary and not sufficient — the diagonal case can land inside $[0, 1]$

3. Average precision is defined with a maximum inside it. State what that maximum repairs, and which earlier lecture proved the problem. **[4]**

Solution

It replaces the non-monotone precision by its running maximum — the non-monotonicity proved in Lecture 3.

- Precision falls at every false positive and can rise again
- Integrating the raw curve would reward the sawtooth rather than the ranking

4. mAP is a mean over classes of a mean over IoU thresholds. Give one thing each averaging hides. **[4]**

Solution

The class mean hides that a one-instance class weighs as much as a 350-instance one; the threshold mean hides the spread between loose and tight matching.

- Unweighted means ignore support
- A single number in the middle stands for two very different regimes

5. A team computes AP per image and averages those. State the direction of the error and its cause. **[4]**

Solution

It is optimistic: single images contain few objects and often score 1.0.

- Averaging many easy sub-problems is not the hard problem
- It is the metric averaged per batch rather than over the set

Lecture 15 Time series

Ch 13

1. For a stationary series, $\mathrm{Var}(X_t - X_{t-h}) = 2\gamma(0)(1 - \rho(h))$. State the condition on $\rho(h)$ under which differencing *increases* the variance. **[4]**

Solution

$\rho(h) < 1/2$.

- The factor $2(1-\rho)$ exceeds 1 exactly then
- So the reflex 'it is not stationary, difference it' can make the problem harder

2. Explain why the seasonal-naive forecast is hard to beat on daily ridership, in terms of the autocorrelation function. **[4]**

Solution

$\rho(7)$ is high, so the lag-7 difference is close to white noise — and copying last week is the model that assumes it is.

- The weekly cycle carries most of the structure
- What remains after removing it is close to unpredictable

3. State why MAPE is the wrong metric for transit ridership, using a specific day as the example. **[4]**

Solution

It divides by the truth, and holiday ridership is a fraction of a working day's, so the same absolute miss on Christmas counts several times what it counts on an ordinary weekday.

- The metric would spend capacity on the days nobody staffs for
- MAE in boardings is in the units the operations team already uses

4. A cross-validated forecast uses `KFold(shuffle=True)` and reports a good score. Name the two conditions the split violates. **[5]**

Solution

No training row may come after a test row, and none may be adjacent to one.

- Shuffling puts a point's own future in the training set
- Neighbouring days are nearly the same number, so an adjacent row nearly gives away the answer

5. In March 2020 the series level falls by three quarters and the model stops working. State whether this is a leak, a bug, or neither, and what it implies for deployment. **[4]**

Solution

Neither — the protocol was correct and the world changed. It implies monitoring, with a written trigger.

- No split protects against a regime change
- A model never re-measured after deployment is an assumption wearing a number's clothes

Lecture 16 Recurrent networks

Ch 13

1. A recurrent network is trained on 56-day windows drawn from one series. Explain why shuffling the *windows* is legitimate while shuffling the *series* is not. **[4]**

Solution

A window is a training example; the series is the thing the example is cut from.

- The order inside a window is the signal, and shuffling windows leaves it intact
- Shuffling the series destroys the very structure the model exists to read

2. The head of the recurrent model is a linear layer on the final hidden state. State why the hidden state alone will not do. **[4]**

Solution

tanh bounds it to $(-1, 1)$ and ridership is not bounded.

- A bounded output cannot reach the target range
- Scaling by a million makes it half-work, which is worse than failing

3. Gates were introduced after a plain recurrent cell failed over long windows. State the mechanism, in terms of what happens to the same matrix over 56 steps. **[4]**

Solution

The same matrix is applied 56 times, so its spectrum is raised to the 56th power; gates provide a path whose derivative is near one.

- Repeated multiplication is the vanishing-gradient argument of Lecture 11 in a new place
- A gate lets information pass without being multiplied at every step

4. A colleague reports a large improvement on the ridership forecast after making the recurrent layer bidirectional. State what has happened, and the one question that settles it. **[5]**

Solution

It is leakage: a bidirectional layer reads the future. Ask whether the whole sequence is available at prediction time.

- A second layer run backwards lets every position see values that have not happened when the forecast is made
- It is Lecture 15's random split wearing a different hat — the number goes up and the system cannot be deployed

5. Report the training MAE beside the test MAE. Name the two failures the pair distinguishes, and say why the test column alone cannot. **[4]**

Solution

Underfitting, where both are poor, and overfitting, where train is far better. They have opposite fixes.

- A single bad test number is consistent with either
- More capacity fixes one and makes the other worse

Lecture 17 Text

Ch 14

1. Show that softmax is invariant to adding a constant to every logit, and state the numerical reason the implementation subtracts the maximum anyway. **[5]**

Solution

e^c cancels between numerator and denominator. Subtracting the max keeps every exponential below 1, and float32 overflows past about 88.7.

- The function ignores the shift; the arithmetic does not
- A confident network reaching logits of 100 is completely ordinary, and e^{100} is not representable — the sum becomes infinite and the ratio a NaN

2. Derive $\partial L / \partial \mathbf{z} = \mathbf{p} - \mathbf{y}$ for cross-entropy on a one-hot target, and state what the row sum of that gradient must be. **[5]**

Solution

$L = -z_c + \log \sum_j e^{z_j}$, whose derivative is $\mathbf{p} - \mathbf{y}$; each row sums to zero.

- The exponential of the true class cancels, so no chain rule through the softmax is needed
- A zero row sum is the derivative form of the shift invariance

3. A loss function is given probabilities rather than logits. State the two failure modes, and which one ships. **[5]**

Solution

Overflow to inf or nan, and a finite but wrong value. The quiet one ships.

- Taking the log of an underflowed probability is the loud failure
- A row whose true class has small probability loses precision silently

4. A model summarises a padded batch at `out[:, -1, :]`. State what that position holds for most reviews, and the invariance test that detects it. **[5]**

Solution

Padding. Encode the same review with two different amounts of padding and require the logits to agree.

- That position is real content only for the longest reviews
- An output-shape test cannot see this; an invariance test can

5. Enlarging a word vocabulary drives the share of unseen distinct test words down, but even the full 79,003-word vocabulary leaves 42.8% of them unseen. Explain why a larger vocabulary is not the fix, and name what is. **[4]**

Solution

A word vocabulary is closed and language is not. Subword tokenisation.

- Two words in five are seen exactly once, so more capacity buys mostly those — and the next review can still contain a word nobody has written yet
- Starting from characters covers every string by construction, so the floor is zero rather than 42.8%

Lecture 18 Attention and transformers

Ch 14–15

1. Scaled dot-product attention divides by $\sqrt{d_k}$. State what the variance of the unscaled dot product is, and what the softmax does without the division. **[5]**

Solution

$\mathrm{Var}(s) = d_k$, so s has standard deviation $\sqrt{d_k}$; without the division the softmax saturates and the gradient vanishes.

- A sum of d_k independent zero-mean products, each of variance 1, has variance d_k — at $d_k = 64$ typical scores are already of order 8
- Large logits make the distribution nearly one-hot, so almost no gradient flows

2. A decoder must not attend to positions after the one it is predicting. State the mechanism that enforces it, and what would leak without it. **[5]**

Solution

A mask applied to the attention scores before the softmax, setting future positions to $-\infty$. Without it the model sees the token it is being asked to predict.

- Attention is all-to-all by construction, so nothing in the mechanism itself respects order
- The training loss would fall and the model would be useless at generation time, when the future genuinely does not exist

3. A pretrained body is fine-tuned at 10^{-3}, the rate that worked for the from-scratch model. Predict what happens, and state the rate the lecture uses instead. **[4]**

Solution

The loss rises and does not recover — steps that size destroy what was pretrained. The lecture fine-tunes at 2×10^{-5}.

- A pretrained block already encodes something; a step tuned for random initialisation is far too large for it
- Fifty times smaller than the 10^{-3} that trained the model from scratch — a step tuned for random initialisation is far too large for a block that already encodes something

4. A vectoriser is fitted before the split on a 400-document corpus. The leaky vocabulary is measurably larger — 60,000 columns against 54,074. State what that buys the model, and why. **[4]**

Solution

Nothing measurable: 0.30 points against a seed spread of 2.53%, with the leak ahead on 10 of 20 seeds. A null result.

- A term occurring only in test documents has an all-zero column in the training rows, so logistic regression gives it coefficient exactly zero — there is no gradient to move it
- What actually leaks is the inverse document frequencies, and an idf is an average over the corpus: the leak changes the scale of features the model already had, and a regularised linear model is nearly indifferent to that

5. A corpus contains duplicate documents across the train/test split. State why no pipeline fixes it, and name the splitter that does. **[4]**

Solution

Nothing was fitted wrongly — the rows were separated wrongly. Use a grouped split.

- `fit` and `transform` were used correctly throughout
- The repair is to keep both copies of an entry on the same side

Lecture 19 Information retrieval: the lexical foundation

Lecture notes

1. A ranking has relevant documents at ranks 1, 3 and 10, with three relevant documents in total. Compute AP and NDCG@10, showing the sums. **[6]**

Solution

AP = $\frac{1}{3}(1 + \frac{2}{3} + \frac{3}{10}) = 0.6556$; NDCG@10 = $\frac{1+0.5+0.2891}{1+0.6309+0.5} = 0.8396$.

- AP averages the precision at each hit and divides by $|R|$, not by the number of hits found
- DCG discounts by $1/\log_2(i+1)$, and the ideal puts all three at ranks 1–3

2. State why $\log_2(i+1)$ is used rather than $\log_2(i)$ in DCG, and whether the choice of discount is derived or conventional. **[4]**

Solution

$\log_2(1) = 0$ would divide by zero at rank 1. The discount is conventional.

- The +1 is a necessity, not a preference
- The shape encodes a belief about attention, and reporting NDCG adopts it

3. BM25 has three parts. Name the failure of raw term counting each one answers, then say which part the ablation prices on its own as costliest, and which part it cannot price alone. **[5]**

Solution

idf answers common words dominating; saturation answers repetition; length normalisation answers long documents. Of those removed singly idf costs most; saturation is never removed on its own.

- Removing idf alone costs 0.1261 NDCG@10 and removing length normalisation alone costs 0.0120 — those are the two the ablation isolates
- The remaining row drops saturation and length together for 0.1809, and a measurement that removes two things at once cannot attribute the loss to either of them

4. 98% of claims match no document under Boolean conjunction. State the property of the conjunction that causes it, and what the field does instead. **[4]**

Solution

One missing term excludes the document entirely. Score rather than filter.

- The conjunction is brittle, and an empty result looks like 'no such document exists'
- Scoring makes a missing term cost something rather than everything

5. Relevance judgements are made by pooling. State what happens to a document no pooled system ranked highly, and the consequence for a genuinely new method. **[4]**

Solution

It is unjudged and scored as irrelevant, so a new method is penalised for finding it.

- Everything outside the pool is treated as not relevant by convention
- Gains on a mature benchmark are partly gains at matching the pool

Lecture 20 Information retrieval: dense retrieval

Lecture notes

1. Explain why a bi-encoder can be indexed and a cross-encoder cannot, and say what the two differences have in common as a cause. **[5]**

Solution

$E(d)$ does not depend on the query, so it can be precomputed; the cross-encoder reads the pair jointly. The same fact causes the accuracy difference.

- Whether the query is present when the document is read decides both
- Nothing can be stored for a function that does not exist until the query arrives

2. A first stage has recall@100 of 0.885. State the highest recall a perfect re-ranker over those 100 candidates can reach, and what follows for where to spend effort. **[4]**

Solution

0.885. Improving the first stage raises the ceiling; improving the second never does.

- A re-ranker reorders a fixed candidate set and cannot retrieve
- 11.5% of relevant documents are simply absent from the candidates

3. In-batch negatives are described as free. State precisely what is free and what is not, and why the batch size is a modelling decision rather than a memory setting. **[5]**

Solution

The B^2 scores are nearly free; the $2B$ encoder passes are not. The batch size is the number of negatives each query is scored against, so it changes the task.

- Encoding grows linearly in B and the dot products quadratically, so the scores cost almost nothing beside the encoder
- Doubling the batch doubles the negatives per query, which is a choice about the problem rather than about the hardware

4. Hard negatives are mined from a first stage's top results. State the trap, and why it is worse at training time than at evaluation time. **[5]**

Solution

Many are relevant but unjudged, so training teaches the model that a correct answer is wrong.

- At evaluation an unjudged document is an accounting error
- At training it is a gradient pushing the model away from the right answer

5. An approximate index reports recall 0.80. State what that number must be measured against, and what is conflated if it is not. **[4]**

Solution

Against the exact scan, never against the relevance judgements.

- Otherwise 'my index missed it' and 'my model ranked it low' become one number
- They have different fixes, so they must be measured separately

Lecture 21 Recommender systems: from ratings to factors

Lecture notes

1. State why the truncated SVD cannot be applied directly to a ratings matrix, naming the condition of Eckart–Young that fails. **[5]**

Solution

The Frobenius norm sums over every entry, and 95.5% of them have no value.

- The theorem is about one fully specified matrix
- Filling the gaps changes the problem into a different one with a different answer

2. A rank-32 SVD of the zero-filled matrix scores worse than predicting one number for everybody. Explain how the optimal approximation can lose to a constant. **[5]**

Solution

It is optimal for the matrix it was given, and that matrix is mostly zeros.

- Ratings run from 1 to 5, so a zero is off the scale rather than low
- The approximation spends its capacity reproducing an assumption nobody made deliberately

3. Write the ALS half-step for one user, and identify it as a familiar estimator. **[5]**

Solution

$p_u = (Q_u^\top Q_u + \lambda I)^{-1} Q_u^\top r_u$ — a ridge regression.

- Q_u contains only the items that user rated, so the missing entries never enter the system
- The system is $d \times d$ however many items the user rated

4. For any invertible M, $(PM)(QM^{-\top})^\top = PQ^\top$. State what this settles about interpreting a latent factor. **[4]**

Solution

The predictions are determined and the factors are not, so an individual dimension has no meaning.

- The objective cannot distinguish P from PM
- A factor plot claims something the model is not claiming

5. Implicit feedback has no negatives. State why an objective over observed entries alone fails, and name the two standard responses. **[5]**

Solution

It is satisfied by predicting yes everywhere. Weighted least squares over all cells, or a pairwise ranking objective.

- Only positives exist, so there is nothing to push down
- The first keeps a closed form and needs a confidence function; the second models only differences

Lecture 22 Recommender systems: neural, and evaluated honestly

Lecture notes

1. A recommender is scored by RMSE on held-out ratings. State why that is the wrong metric, using the set the model actually operates on. **[5]**

Solution

RMSE is computed on films the user chose to watch and rate; a recommender operates on the complement.

- It weights every prediction equally when only ten items are shown
- A constant offset ruins RMSE and changes no ranking at all

2. The same model scores HR@10 of 0.0926 and 0.8102 on the same data. Name the two decisions that differ, and which moves the number more. **[5]**

Solution

Random against temporal split, and sampled-100 against the full catalogue. Sampling moves it far more.

- Ten of 101 candidates is already a tenth of the set
- A random ranker goes from 0.0027 to 0.0990 under the same change

3. Explain why a sampled metric is not merely noisier than a full-catalogue one, and give the consequence for comparing two published systems. **[5]**

Solution

The 100 negatives are drawn uniformly from a mostly-tail catalogue, so the distortion depends on the model. Orderings can reverse.

- A popularity model competes against almost no popular items
- A uniform bias would at least preserve rankings; this one does not

4. A sampled softmax draws negatives in proportion to popularity. State the correction, and what enters the model if it is skipped. **[4]**

Solution

Subtract $\log q(j)$ from each sampled negative's score. Without it, popularity bias enters the gradient.

- The sample is not the catalogue, and the estimator is biased without reweighting
- The model then systematically under-recommends popular items

5. A recommender is trained on interactions produced by an earlier recommender. Name the effect, and the earlier lecture whose problem it repeats. **[4]**

Solution

The feedback loop — the pooling problem of Lecture 19.

- The log records what a previous system chose to show
- A model that would have shown something better scores badly, because the evidence was never collected

Lecture 23 Vision transformers and multimodal retrieval

Ch 15–16

1. Two encoders are trained separately, one on images and one on text. Explain why their cosine similarity carries no information about matching, using a rotation argument. **[5]**

Solution

Any rotation of the image space leaves every image–image cosine and the image loss unchanged, while changing every image–text cosine.

- The objective cannot distinguish the rotated space from the original
- So it did not pick one, and the relative geometry is arbitrary

2. In d dimensions two independent random unit vectors have a cosine with standard deviation $1/\sqrt{d}$. State what an unrelated pair looks like at $d = 512$, and why -1 is the wrong intuition. **[5]**

Solution

Near zero, with a spread of about 0.044. There is exactly one point at cosine -1, so most pairs cannot be there.

- Unrelated means orthogonal in high dimensions, not opposite
- It is why a contrastive loss targets zero for a non-matching pair

3. A contrastive loss is computed at $\tau = 1$ on cosine scores. State why it barely trains, and what τ changes and does not change. **[5]**

Solution

Cosines span at most 2, so the softmax is nearly uniform. τ changes where the gradient goes, not which column is largest.

- A range of 2 gives a ratio of at most e^2 across all competitors
- Temperature cannot change the accuracy of a fixed similarity matrix

4. Recall@10 is reported over 200 candidates. State the exact random baseline, and why it is computed rather than simulated. **[3]**

Solution

$10/200 = 5\%$.

- One relevant item ranked uniformly gives $P(\text{rank} \le k) = k/n$
- A closed form needs no seed and carries no noise

5. An entry with no written description scores zero on a text retrieval route. State whether this is a ranking failure or a structural one, and what follows. **[4]**

Solution

Structural — an entry with no text is not in a text index at all.

- No amount of re-ranking reaches a document that was never indexed
- The repair is another route, not a better scorer

Lecture 24 Generation, retrieval-augmented systems, and where this leaves you

Ch 15–16

1. A contrastive loss is written as `img @ txt.T / tau` on the output of `get_image_features`. State the defect, and the assertion that would have caught it. **[5]**

Solution

The features are not normalised, so the product is not a cosine. Assert every row has unit norm.

- The entries carry both magnitudes, so the temperature divides a quantity with no fixed scale
- The norms vary by 1.32× across this catalogue for no semantic reason — a vector's length is whatever the last linear layer happened to scale it to — so an unnormalised score ranks partly by how loudly an encoder spoke

2. An auto-caption is generated for catalogue entries that have no description. State what it recovers, and the failure mode nothing in the evaluation detects. **[5]**

Solution

It recovers part of the lost recall. A caption that is wrong makes an entry findable under the wrong query.

- Scoring generated captions against human captions of the same image is a friendly test
- Worse than unfindable, and no metric here sees it

3. A retrieval-augmented system is measured by whether cited stock numbers exist. State what that measures, and what it does not. **[4]**

Solution

It measures grounding, not helpfulness.

- A cited entry can exist and still be a bad recommendation
- It is the cheap half, and saying which half was measured is the point

4. In a retrieval-augmented pipeline the retriever's recall bounds the whole system. Explain why, and what that implies about where to invest. **[4]**

Solution

If the right entry is not retrieved, no generation recovers it.

- The generator can only work with what it is given
- The first stage sets the ceiling, exactly as in Lecture 20

5. Give the five rules the course ends on, and state the property the deck points out that all five share. **[5]**

Solution

Split before anything is fitted; compute the trivial baseline first; two numbers measured on different rows are not comparable; a difference smaller than the spread is not a difference; change one thing. None of them is mathematics.

- Every one governs what a number may be compared with, not how a model is fitted — which is why none of them needed any
- They are the five that would have caught every row of the catalogue of silent failures, each of which runs to completion and raises nothing